Final Exam

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Overall Evaluation - Thoughts

Left To-Do

- AI Comparisons/Evaluations — Additional Questions

Short Answer Questions

1.

In lecture, we discussed an example taken from the book *Generalized Linear Models* by McCullagh and Nelder that illustrated the use of basic quasi-likelihood. In that example, use was made of Pearson residuals,

$$r_i = \frac{(y_i - \hat{\mu}_i)}{[\text{var}(\hat{\mu}_i)]^{1/2}}.$$

Why are Pearson residuals better suited for use with quasi-likelihood than are deviance residuals?

Answer - Me

Pearson residuals are better suited for quasi-likelihood diagnostics because quasi-likelihood specifies only the first two moments of the response — $E(Y_i) = \mu_i$ and $\text{var}(Y_i) = (1/\phi)V(\mu_i)$ — and Pearson residuals are defined entirely in terms of those moments. Deviance residuals, by contrast, require a fully specified log-likelihood to construct the deviance, which is precisely what quasi-likelihood forgoes. In that sense, deviance residuals are not naturally available in the quasi-likelihood framework.

There is also a deeper inferential point. The asymptotic theory underlying deviance-based diagnostics relies on large-sample chi-squared approximations justified through a fully specified likelihood. Pearson residuals carry no such requirement; their moment-based construction means their behavior is governed directly by the variance function $V(\mu_i)$, which is the central modeling object in quasi-likelihood. Their use is therefore on firmer (exact-theory) footing, making them the more appropriate tool for assessing whether the specified $V(\mu_i)$ adequately describes the observed variability.

Answer - AI

Pearson residuals are better suited for quasi-likelihood because **quasi-likelihood methods do not fully specify a probability model**, and Pearson residuals depend only on the **mean–variance relationship**, which is exactly what quasi-likelihood keeps.

Evaluation

AI correctly identifies that Pearson residuals require only the mean–variance relationship, which is all quasi-likelihood specifies, but the answer stops there. In this sense, I would regard it as incomplete.

2.

In any problem involving censored observations for which we will use a likelihood that contains terms for left censored observations and/or interval censored observations and/or right censored observations, what is a major assumption that affects the validity of the analysis?

Answer - Me

That major assumption is that the censoring mechanism is independent of the mechanism producing observed values, and the Notes go on to say, *"which is similar to an assumption of missing at random or, in a situation lacking covariates, missing completely at random."*

Answer - AI

A key assumption in likelihood-based analysis with left/right/interval censored data is:

> **The censoring mechanism is non-informative (a.k.a. independent or ignorable) given the model parameters.**

Evaluation

The AI answer correctly identifies the core assumption, but lacks additional context I believe is meaningful (not just because of how it was explicitly noted in the course text).

3.

There is some evidence that Fisher may have considered aspects of a testing problem such as normality, constant variance, and perhaps even independence as part of the hypothesis to be tested. This would not be possible under a Neyman-Pearson framework in which these aspects of the test must be given as assumptions under which the test is valid. Why?

Answer - Me

The core issue is decision-based: the Neyman-Pearson framework is built around choosing either H_0 or H_1. If distributional assumptions (normality, constant variance, independence) are folded into H_0, rejecting H_0 no longer points toward a specific, actionable alternative. As the text notes, it becomes unclear how one should *"behave under a decision that a null hypothesis is not entirely true, but any one of an array of other possibilities might be."* The decision rule collapses because H_0 and H_1 no longer constitute a **logical disjunction**, i.e., they no longer partition Θ, which is the foundational structural requirement of the Neyman-Pearson setup.

This non-specificity also destroys the power machinery. Type II error and power are defined relative to a *specific* alternative: to compute $\beta(\theta)$, one needs the distribution of the test statistic under H_1, which requires a fully specified model. If the alternative is effectively "anything other than a normal distribution with equal variances and independent errors," that reference distribution is undefined. Without it, power calculations are impossible, sample size cannot be determined, and no claim to optimality — UMP or otherwise — can be made.

Answer - AI

The short answer is: the Neyman–Pearson framework requires a **fully specified probabilistic model before you even define the test**, whereas Ronald Fisher treated model features more flexibly—as things that could themselves be scrutinized by the data.

In the Neyman–Pearson framework, assumptions like normality and independence are **built into the mathematical foundation of the test**. If you tried to test them within the same framework, you would lose the very distributional results needed to control error rates—so the procedure would no longer be valid.

Evaluation

AI gets part of the answer, but misses some key details, namely that H_0 and H_1 must partition Θ (violated when assumptions are folded into H_0), and that power requires a fully specified alternative (which is impossible, as noted in my own answer.)

4.

What is the purpose of using what in class was called a comparison function $h(\theta, \hat{\theta}_n)$ in a parametric bootstrap?

Answer - Me

The comparison function $h(\theta, \hat{\theta}_n)$ (referred to in the Notes as a discrepancy function) serves to reduce or eliminate the dependence of the sampling distribution of $\hat{\theta}_n$ on the unknown true parameter θ. Because the sampling distribution of $\hat{\theta}_n$ itself depends on θ, the bootstrap instead targets the distribution of $h(\theta, \hat{\theta}_n)$. The ideal case is a **pivotal quantity**, where the distribution of $h(\theta, \hat{\theta}_n)$ is entirely free of θ, making the bootstrap approximation as stable and reliable as possible. The choice of h is flexible, and different choices characterize different aspects of the estimator's sampling behavior, leading to different bootstrap interval procedures. It is worth noting that the bootstrap is approximating the distribution of $h(\theta, \hat{\theta}_n)$, not the distribution of $\hat{\theta}_n$ itself, a distinction the Notes treat as essential.

Answer - AI

It lets you replace the unknown distribution:

$$h(\theta, \hat{\theta}_n)$$

with the computable bootstrap approximation:

$$h(\hat{\theta}_n, \hat{\theta}_n^*).$$

So instead of trying to approximate the distribution of $(\hat{\theta}_n)$ directly, you approximate a **transformed version** that behaves the same in both worlds.

Evaluation

AI identifies the *what*, but lacks detail on the *why*.

5.

Why are normed profile likelihoods of no help in finding maximum likelihood estimates?

Answer - Me

In a word: Because calculating a normed profile likelihood requires the maximum likelihood estimate to calculate.

Per the Notes: normed profile likelihoods require the unrestricted joint MLE $\hat{\boldsymbol{\theta}}_n = (\hat{\boldsymbol{\theta}}_1, \hat{\boldsymbol{\theta}}_2)$ in the denominator of the equation (the expression of the *normed profile likelihood* for θ_1):

$$R_n^p(\theta_1) = \max_{\boldsymbol{\theta}_2} \left[\frac{L_n(\theta_1, \boldsymbol{\theta}_2)}{L_n(\hat{\boldsymbol{\theta}}_n)} \right]$$

Answer - AI

The crucial point:

$$\arg\max_{\theta} \tilde{L}_p(\theta) = \arg\max_{\theta} L_p(\theta).$$

Multiplying or dividing by a **positive constant** does not change where the maximum occurs.

So:

- The MLE is determined by where the likelihood is largest.
- Normalizing only changes the *scale*, not the *location* of the maximum.

Evaluation

AI does not directly answer the question, and is simply wrong for that.

6.

Assessment of a Bayesian analysis is often approached through use of the posterior predictive distribution for new observations,

$$p(\mathbf{y}^0 \mid \mathbf{y}) = \int f(\mathbf{y}^0 \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathbf{y})\, d\boldsymbol{\theta}.$$

(a)

For this definition to hold, $\mathbf{Y}^0$ must follow the same distribution as $\mathbf{Y}$, depending on the same value of $\boldsymbol{\theta}$. What else must be true about the relation between $\mathbf{Y}^0$ and $\mathbf{Y}$ for this definition of a posterior predictive distribution to be valid?

Answer - Me

The other thing that must hold is $\mathbf{Y}^0$ and $\mathbf{Y}$ must be **conditionally independent given $\boldsymbol{\theta}$**. This allows the factorization $f(\mathbf{y}^0, \mathbf{y} \mid \boldsymbol{\theta}) = f(\mathbf{y}^0 \mid \boldsymbol{\theta})\, f(\mathbf{y} \mid \boldsymbol{\theta})$, which is necessary to reduce the joint predictive to the integral form shown.

Answer - AI

In addition to ($\mathbf{Y}^0$) and ($\mathbf{Y}$) sharing the same sampling model conditional on ($\boldsymbol{\theta}$), a crucial requirement is:

> **Conditional independence given the parameter ($\boldsymbol{\theta}$).**

Formally, we must have:

$$\mathbf{Y}^0 \perp \mathbf{Y} \mid \boldsymbol{\theta}.$$

Evaluation

AI answer is correct and complete, missing only a minor detail (the factorization) that I included in my answer.

(b)

There are two other predictive distributions that can be useful in a Bayesian analysis. One can be used with the standard data model and prior setup consisting of a data model $f(\mathbf{y}|\boldsymbol{\theta})$ and a prior $\pi(\boldsymbol{\theta})$. The other is connected specifically with hierarchical models consisting of $f(\mathbf{y}|\boldsymbol{\theta})$, $g(\boldsymbol{\theta}|\boldsymbol{\lambda})$ and $\pi(\boldsymbol{\lambda})$. Define these predictive distributions, and indicate their primary uses.

Answer - Me

The first is the **prior predictive distribution**, applicable under a standard data model and prior $f(\mathbf{y} \mid \boldsymbol{\theta})$, $\pi(\boldsymbol{\theta})$:

$$p(\mathbf{y}) = \int f(\mathbf{y} \mid \boldsymbol{\theta})\, \pi(\boldsymbol{\theta})\, d\boldsymbol{\theta}.$$

Its primary uses are prior predictive model checking, i.e., assessing whether the model implies plausible data before observing $\mathbf{y}$, in addition to being the marginal likelihood in Bayes factor model comparisons.

The second is the **posterior predictive distribution for new random parameters $\boldsymbol{\theta}^*$**, connected specifically with hierarchical models $f(\mathbf{y} \mid \boldsymbol{\theta})$, $g(\boldsymbol{\theta} \mid \boldsymbol{\lambda})$, $\pi(\boldsymbol{\lambda})$:

$$p(\boldsymbol{\theta}^* \mid \mathbf{y}) = \int g(\boldsymbol{\theta}^* \mid \boldsymbol{\lambda})\, p(\boldsymbol{\lambda} \mid \mathbf{y})\, d\boldsymbol{\lambda}.$$

Its primary use is making probability statements about what will occur in new, unobserved situations drawn from the same mechanism, e.g., predicting the economic optimal nitrogen rate in a future trial (as noted in the Notes). It is preferred over evaluating $g(\boldsymbol{\theta}^* \mid \hat{\boldsymbol{\lambda}})$ at a point estimate because it propagates full posterior uncertainty in $\boldsymbol{\lambda}$.

Answer - AI

1. Prior predictive distribution (marginal predictive)

Definition Given a standard Bayesian model:

- likelihood: ($f(\mathbf{y} \mid \boldsymbol{\theta})$)
- prior: ($\pi(\boldsymbol{\theta})$)

the **prior predictive distribution** is:

$$p(\mathbf{y}) = \int f(\mathbf{y} \mid \boldsymbol{\theta}), \pi(\boldsymbol{\theta}), d\boldsymbol{\theta}.$$

Interpretation This is the distribution of the data **before seeing any observations**, integrating over prior uncertainty in $(\boldsymbol{\theta})$.

Primary uses

1. **Model checking (prior-to-data coherence)**
   - Do simulated datasets from the model "look like" plausible data?
   - Helps detect whether the prior + likelihood combination is unreasonable.
2. **Design / prior sensitivity assessment**
   - Shows what kinds of data the model expects a priori.
   - Useful for diagnosing overly informative or implausible priors.
3. **Bayesian model comparison (marginal likelihood)**
   - This is the key quantity in Bayes factors:

$$p(\mathbf{y} \mid M)$$

2. Prior predictive for hierarchical models (conditional predictive at the hyperparameter level)
Now consider a hierarchical model:

- $(f(\mathbf{y} \mid \boldsymbol{\theta}))$
- $(g(\boldsymbol{\theta} \mid \boldsymbol{\lambda}))$
- $(\pi(\boldsymbol{\lambda}))$

Definition The **hierarchical prior predictive distribution** is:

$$p(\mathbf{y}) = \int \int f(\mathbf{y} \mid \boldsymbol{\theta}), g(\boldsymbol{\theta} \mid \boldsymbol{\lambda}), \pi(\boldsymbol{\lambda}), d\boldsymbol{\theta}, d\boldsymbol{\lambda}.$$

Equivalently:

$$p(\mathbf{y}) = \int f(\mathbf{y} \mid \boldsymbol{\lambda}), \pi(\boldsymbol{\lambda}), d\boldsymbol{\lambda}, \quad \text{where } f(\mathbf{y} \mid \boldsymbol{\lambda}) = \int f(\mathbf{y} \mid \boldsymbol{\theta}) g(\boldsymbol{\theta} \mid \boldsymbol{\lambda}) d\boldsymbol{\theta}.$$

Interpretation This is the distribution of data **before observing anything**, but now accounting for both:

- within-group variation $((\boldsymbol{\theta} \mid \boldsymbol{\lambda}))$
- between-group variation $((\boldsymbol{\lambda}))$

Primary uses

1. **Hierarchical model checking**
   - Does the full multilevel model generate realistic data?
   - Important for checking whether shrinkage assumptions are plausible.
2. **Assessing prior structure at multiple levels**
   - Evaluates whether hyperpriors imply reasonable variability across groups.
3. **Understanding induced marginal behavior**
   - Shows what marginal distributions the hierarchy implies for observables (often heavy-tailed or overdispersed forms).

Evaluation

AI is correct w/r/t the prior predictive distribution, in toto, but gets the second distribution wrong entirely. In that sense, the AI answers a different question and is simply wrong for the second part.

Additional Questions

1.

A set of environmental samples (say soil samples) have been taken to determine the level of contamination of a site by polychlorinated biphenyls (PCBs). Two laboratories were used to analyze the samples, with n_1 samples being sent to Lab 1 and n_2 samples sent to Lab 2. Lab 1 has a detection limit of 0.01 μg/L, while Lab 2 has a detection limit of 0.03 μg/L. Both labs have a precision of 0.005 μg/L for values reported to two decimal places. The data recorded are laboratory (1 or 2), sample number, concentration in μg/L, and an indicator as to whether the sample was below detection limits (1) or not (0). We have decided to model concentrations with a gamma distribution, parameterized for a random variable X and parameters $\alpha > 0$ and $\beta > 0$ as,

$$f(x \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x); \quad x > 0.$$

Write the likelihood function for the entire collection of n_1 samples analyzed at Lab 1 and n_2 samples analyzed at Lab 2. Assume independence of samples, and note that our objective is to estimate the distribution of concentrations, NOT to compare the results between the two laboratories.

HINT: Define random variables and set up your indexing system first.

Answer - Me

Let $i \in \{1, 2\}$ index the laboratory and $j = 1, \ldots, n_i$ index the sample within lab i, where n_1 and n_2 denote the number of samples sent to Lab 1 and Lab 2, respectively.

Let X_{ij} denote the true PCB concentration of sample j analyzed at lab i, with

$$X_{ij} \overset{\text{iid}}{\sim} \text{Gamma}(\alpha, \beta), \quad i = 1, 2, \quad j = 1, \ldots, n_i.$$

Each sample is associated with an indicator

$$\delta_{ij} = \begin{cases} 1, & X_{ij} < c_i \quad \text{(below detection limit)} \\ 0, & X_{ij} \geq c_i \quad \text{(detected)} \end{cases}$$

and, when $\delta_{ij} = 0$, a reported concentration y_{ij} recorded to precision $\Delta = 0.005$ μg/L. Define the index sets

$$\mathcal{D}_i = \{j : \delta_{ij} = 0\}, \quad \mathcal{U}_i = \{j : \delta_{ij} = 1\}, \quad |\mathcal{D}_i| + |\mathcal{U}_i| = n_i.$$

Assuming independence across all samples, the likelihood factors as a product over both labs and all samples within each lab. Each observation contributes a probability mass under the Gamma model corresponding to its censoring type: detected samples contribute an interval probability of width 2Δ, while below-detection samples contribute the probability of falling below the lab's detection limit c_i.

$$L(\alpha, \beta) = \prod_{i=1}^{2} \left[\prod_{j \in \mathcal{D}_i} \Big[F(y_{ij} + \Delta \mid \alpha, \beta) - F(y_{ij} - \Delta \mid \alpha, \beta) \Big] \prod_{j \in \mathcal{U}_i} F(c_i \mid \alpha, \beta) \right]$$

where $F(\cdot \mid \alpha, \beta)$ is the Gamma CDF, $\Delta = 0.005$ μg/L is the common reporting precision, and $c_1 = 0.01$, $c_2 = 0.03$ μg/L are the detection limits for Labs 1 and 2, respectively. Note that $F(0 \mid \alpha, \beta) = 0$ for the Gamma distribution, so the left-censored contribution simplifies directly to $F(c_i \mid \alpha, \beta)$. The shared parameters (α, β) across both labs reflect our objective of estimating a single underlying concentration distribution.

Answer - AI

Let (X_{ij}) denote the true PCB concentration for sample (j) analyzed at lab (i), where

- $(i = 1, 2)$ indexes the laboratory,
- $(j = 1, \ldots, n_i)$,
- and $(X_{ij} \overset{\text{ind}}{\sim} \text{Gamma}(\alpha, \beta))$ with density

$$f(x \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0.$$

Let $(c_1 = 0.01)$ and $(c_2 = 0.03)$ be the detection limits for Lab 1 and Lab 2 respectively.

Define an indicator:

$$\delta_{ij} = \begin{cases} 1, & X_{ij} < c_i \quad \text{(below detection limit)} \\ 0, & X_{ij} \geq c_i \quad \text{(detected)} \end{cases}$$

We observe:

- If $(\delta_{ij} = 0)$: we observe the actual value (x_{ij}).
- If $(\delta_{ij} = 1)$: we only know $(X_{ij} < c_i)$, so we observe a censored contribution.

Likelihood contribution for one observation

(i) Detected observation $((\delta_{ij} = 0))$

$$L_{ij} = f(x_{ij} \mid \alpha, \beta)$$

(ii) Non-detect $((\delta_{ij} = 1))$

$$L_{ij} = P(X_{ij} < c_i) = \int_0^{c_i} f(x \mid \alpha, \beta), dx = F(c_i \mid \alpha, \beta)$$

where $(F(\cdot))$ is the Gamma CDF.

Assuming independence across all samples (and across labs), the likelihood is:

$$L(\alpha, \beta) = \prod_{i=1}^{2} \prod_{j=1}^{n_i} \left[f(x_{ij} \mid \alpha, \beta)^{1-\delta_{ij}}; F(c_i \mid \alpha, \beta)^{\delta_{ij}} \right].$$

Substituting the gamma density:

$$L(\alpha, \beta) = \prod_{i=1}^{2} \prod_{j=1}^{n_i} \left[\left(\frac{\beta^\alpha}{\Gamma(\alpha)} x_{ij}^{\alpha-1} e^{-\beta x_{ij}} \right)^{1-\delta_{ij}} (F(c_i \mid \alpha, \beta))^{\delta_{ij}} \right].$$

Evaluation

2.

Consider four random variables X_1, X_2, Y, and Z. You are told that X_1 and X_2 are conditionally independent given the pair (Y, Z) but that Y and Z are not independent. Determine two conditions, similar in form but one for X_1 and one for X_2, that are sufficient for $[X_1, X_2 \mid Y, Z] = [X_1 \mid Y][X_2 \mid Z]$.

Answer - Me

We begin from the given conditional independence of X_1 and X_2 given (Y, Z):

$$[X_1, X_2 \mid Y, Z] = [X_1 \mid Y, Z]\,[X_2 \mid Y, Z].$$

For this to equal $[X_1 \mid Y][X_2 \mid Z]$, it suffices to impose the following two conditions:

1. $X_1 \perp Z \mid Y$, so that $[X_1 \mid Y, Z] = [X_1 \mid Y]$;
2. $X_2 \perp Y \mid Z$, so that $[X_2 \mid Y, Z] = [X_2 \mid Z]$.

Substituting both into the factorization above yields

$$[X_1, X_2 \mid Y, Z] = [X_1 \mid Y]\,[X_2 \mid Z],$$

as required. Each condition asserts that one X variable is conditionally independent of the "irrelevant" member of the conditioning pair given the other, and the two conditions are symmetric in form.

Answer - AI

The two sufficient conditions are:

$$\boxed{X_1 \perp Z \mid Y} \quad \text{and} \quad \boxed{X_2 \perp Y \mid Z.}$$

Evaluation

3.

Consider a situation in which we have two scientific or substantive processes of interest. These could, for example, be two environmental processes in a rainforest such as tree growth and evapotranspiration, or two constructs in the social sciences such as access to preventative health care and social status in rural areas. We may have some scientific basis to believe these processes are not independent of one another. We also have two observable quantities that we believe are direct reflections of those processes, such as vegetative biomass and respiration in sample plots for the rainforest example, or number of rural medical centers and poverty rates in counties for the social science example.

(a)

First, define some notation to make a problem such as those described (two non-independent processes and two direct indicators) fit into the notational framework of the previous question. That is, define X_1, X_2, Y, and Z in terms of processes (say process 1 and process 2), and observables (say observable 1 and observable 2). You can, without additional notation, assume that writing down one observable random variable implies a collection of values from sampling units.

Answer - Me

Let Y and Z represent Process 1 and Process 2, respectively. These processes are not independent of one another, i.e., $[Y, Z] \neq [Y][Z]$, reflecting the substantive belief that the two underlying processes are related.

Let X_1 and X_2 represent Observable 1 and Observable 2, respectively, where X_1 is a direct reflection of Process 1 (Y) and X_2 is a direct reflection of Process 2 (Z).

Answer - AI

Let the two underlying scientific processes of interest be:

- Process 1: (P_1) (e.g., tree growth rate, or access to healthcare)
- Process 2: (P_2) (e.g., evapotranspiration, or poverty level)

We assume these processes are **not independent**, i.e., ($P_1 \not\perp P_2$).

Now define two observable quantities that are measured from sampling units and serve as proxies or indicators of these processes:

- Observable 1: (O_1), which reflects Process 1
- Observable 2: (O_2), which reflects Process 2

To align with the previous conditional independence framework, define:

- ($X_1 := O_1$) (observable indicator of Process 1)
- ($X_2 := O_2$) (observable indicator of Process 2)
- ($Y := P_1$) (latent underlying Process 1)
- ($Z := P_2$) (latent underlying Process 2)

Under this mapping:

- (X_1) is a direct measurement of Process 1
- (X_2) is a direct measurement of Process 2
- (Y) and (Z) are the true (possibly dependent) scientific processes generating the system
- The observables (X_1, X_2) are assumed to arise from their respective processes (possibly with measurement error or sampling variation)

Thus, the structure is:

$$X_1 \leftarrow Y, \quad X_2 \leftarrow Z, \quad \text{with } Y \not\perp Z.$$

This places the scientific example directly into the probabilistic framework of the earlier question, where the goal is to understand conditional relationships of observables given underlying dependent processes.

Evaluation

(b)

Using an extension of your notation from (a), identify the distributional forms that will be needed to formulate a hierarchical model. For example, if I desire to model the distribution of a random variable W conditional on the value of another random variable S with a density or mass function that has a parameter θ I would say I need to formulate a model for $[W \mid S, \theta]$.

Answer - Me

A hierarchical model for this setting requires three distributional components.

At the **data level**, we need a model for each observable given its directly associated process:

$$[X_1 \mid Y, \boldsymbol{\theta}_1] \quad \text{and} \quad [X_2 \mid Z, \boldsymbol{\theta}_2]$$

where $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ parameterize the relationship between each observable and its respective process.

At the **process level**, because Y and Z are not independent, we require a joint model for the two processes:

$$[Y, Z \mid \boldsymbol{\theta}_3]$$

where $\boldsymbol{\theta}_3$ governs the marginal behavior of each process as well as the dependence between them. The full parameter vector for the hierarchical model is then $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3)$.

Answer - AI

Putting the layers together, the full model requires specifying:

(i) Latent process layer

$$[Y, Z \mid \phi]$$

(ii) Observation layer 1

$$[X_1 \mid Y, \theta_1]$$

(iii) Observation layer 2

$$[X_2 \mid Z, \theta_2]$$

Evaluation

(c)

Suppose we are willing to assume the conditions given and identified in your answer to the previous question. Now identify what part(s) of the model formulation from part (b) is(are) made more simple as a result of that assumption.

Answer - Me

The conditions from the prior question — that $X_1 \perp Z \mid Y$ and $X_2 \perp Y \mid Z$ — affect only the **data level** of the model. Without those conditions, we would in general need to formulate a joint model for the two observables given both processes,

$$[X_1, X_2 \mid Y, Z, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2],$$

which could require specifying a dependence structure between X_1 and X_2 even after conditioning on (Y, Z). Under the stated conditions, this joint model factorizes as

$$[X_1, X_2 \mid Y, Z, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2] = [X_1 \mid Y, \boldsymbol{\theta}_1]\,[X_2 \mid Z, \boldsymbol{\theta}_2],$$

so each observable need only be modeled conditionally on its own directly associated process. The two data-level components from part (b) can therefore be specified independently of one another.

The process-level component $[Y, Z \mid \boldsymbol{\theta}_3]$ is **not** simplified — Y and Z remain non-independent by assumption, so a joint model for the two processes is still required.

Answer - AI

The assumption simplifies the model specifically by:

1. **Forcing factorization of the observation model**

$$[X_1, X_2 \mid Y, Z] \to [X_1 \mid Y], [X_2 \mid Z]$$

2. **Removing any need for a joint measurement error or dependence structure between (X_1) and (X_2)**
3. **Allowing complete separation of observation parameters**

$$\theta \to (\theta_1, \theta_2)$$

4. **Yielding a fully factorized hierarchical likelihood**

$$[Y, Z \mid \phi], [X_1 \mid Y, \theta_1], [X_2 \mid Z, \theta_2].$$

In short: the assumption collapses what could have been a coupled, potentially complex observation model into two conditionally independent measurement models, greatly simplifying both specification and inference.

Evaluation

(d)

In one or possibly two (but one should really do it) sentence or sentences, state what the assumption implies about the actual processes and observable quantities involved in the problem. If you wish, you may use one of the hypothetical examples given in the preamble to this question to make things more concrete (but you certainly do not have to do that).

Answer - Me

The assumption implies that each observable is driven entirely by its own associated process — once the value of that process is known, the other process contributes no additional information about the observable.

Answer - AI

The assumption implies that, conditional on their respective underlying processes, each observable quantity depends only on its own process and carries no additional information about the other process or its measurement, so any association between the two observables is entirely mediated through the dependence between the underlying processes themselves. In other words, once we account for (say) tree growth affecting biomass and evapotranspiration affecting respiration, biomass tells us nothing further about evapotranspiration beyond what is already encoded through the latent ecological linkage between the two processes.

Evaluation

As per the question comment, *"but one should really do it"*, AI really didn't take that to heart.

4.

If you have heard any news about the stock market over the past 10 weeks or so, you know that there have been some days with extreme losses, some with sizable "bounce-back" gains, and some with relatively little movement. This is called high "volatility" in the market. There are all kinds of ways one might model a high volatility situation, but suppose we suggest the following:

Let $\{Y_t : t = 1, \ldots, T\}$ denote random variables associated with the change in an index of the stock market on day t (change from day $t-1$). Formulate the following model

$$Y_t = \mu + \epsilon_t$$

$$\epsilon_t \sim N(0,\, S_t^2)$$

$$S_t = \alpha S_{t-1} + v_t$$

$$v_t \overset{\text{iid}}{\sim} N(0,\, \sigma^2)$$

(a)

Determine the expected values, variances, and one-step covariances of S_t (that is, the covariance of S_t with S_{t-1} or S_{t+1}).

Hint: If you recall them, you may use the results we obtained in class for AR(1) processes without derivation.

Answer - Me

The process governing S_t is

$$S_t = \alpha S_{t-1} + v_t, \quad v_t \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

which is an AR(1) process in S_t. We may therefore apply AR(1) results directly.

Since $E[v_t] = 0$ for all t, iterating expectations gives

$$E[S_t] = 0.$$

Assuming second-order stationarity, which requires $|\alpha| < 1$, the AR(1) variance result gives

$$\text{var}(S_t) = \frac{\sigma^2}{1-\alpha^2}.$$

From the general AR(1) covariance formula $\text{cov}(S_t, S_{t-k}) = \alpha^k \sigma^2/(1-\alpha^2)$, evaluated at lag $k = 1$,

$$\text{cov}(S_t, S_{t-1}) = \frac{\alpha\sigma^2}{1-\alpha^2}.$$

Answer - AI

We recognize that ({S_t}) is an AR(1) process:

$$S_t = \alpha S_{t-1} + v_t, \quad v_t \overset{iid}{\sim} N(0, \sigma^2).$$

Assume ($|\alpha| < 1$) so the process is (weakly) stationary.

Take expectations on both sides:

$$\mathbb{E}[S_t] = \alpha \mathbb{E}[S_{t-1}] + \mathbb{E}[v_t].$$

Since ($\mathbb{E}[v_t] = 0$), stationarity implies ($\mathbb{E}[S_t] = \mathbb{E}[S_{t-1}] = \mu_S$), so:

$$\mu_S = \alpha\mu_S \quad \Rightarrow \quad (1-\alpha)\mu_S = 0 \quad \Rightarrow \quad \mu_S = 0.$$

So,

$$\boxed{\mathbb{E}[S_t] = 0.}$$

Let ($\gamma_0 = \mathrm{Var}(S_t)$). Then:

$$\mathrm{Var}(S_t) = \mathrm{Var}(\alpha S_{t-1} + v_t).$$

Using independence of (S_{t-1}) and (v_t):

$$\gamma_0 = \alpha^2\gamma_0 + \sigma^2.$$

Solve:

$$\gamma_0(1-\alpha^2) = \sigma^2 \quad \Rightarrow \quad \boxed{\mathrm{Var}(S_t) = \frac{\sigma^2}{1-\alpha^2}}.$$

For one-step covariances

(i) $Cov(S_t, S_{t-1})$

$$\mathrm{Cov}(S_t, S_{t-1}) = \mathrm{Cov}(\alpha S_{t-1} + v_t, S_{t-1}) = \alpha\mathrm{Var}(S_{t-1}) + 0$$

So:

$$\boxed{\mathrm{Cov}(S_t, S_{t-1}) = \alpha\frac{\sigma^2}{1-\alpha^2}}.$$

(ii) $Cov(S_t, S_{t+1})$

Use:

$$S_{t+1} = \alpha S_t + v_{t+1}.$$

Then:

$$\mathrm{Cov}(S_t, S_{t+1}) = \mathrm{Cov}(S_t, \alpha S_t + v_{t+1}) = \alpha\mathrm{Var}(S_t).$$

So:

$$\boxed{\mathrm{Cov}(S_t, S_{t+1}) = \alpha\frac{\sigma^2}{1-\alpha^2}}.$$

Evaluation

(b)

Determine the expected values, variances, and one-step covariances of ϵ_t.

Hint: Use of the standard conditioning argument for expected values and variances might be valuable here.

Answer - Me

By the law of total expectation, conditioning on S_t,

$$E[\epsilon_t] = E\big[E[\epsilon_t \mid S_t]\big] = E[0] = 0,$$

since $\epsilon_t \mid S_t \sim N(0, S_t^2)$.

By the law of total variance,

$$\mathrm{var}(\epsilon_t) = E\big[\mathrm{var}(\epsilon_t \mid S_t)\big] + \mathrm{var}\big(E[\epsilon_t \mid S_t]\big) = E[S_t^2] + \mathrm{var}(0) = E[S_t^2].$$

Since $E[S_t^2] = \mathrm{var}(S_t) + (E[S_t])^2 = \sigma^2/(1-\alpha^2) + 0$, we have

$$\mathrm{var}(\epsilon_t) = \frac{\sigma^2}{1-\alpha^2}.$$

Since both marginal means are zero, $\mathrm{cov}(\epsilon_t, \epsilon_{t-1}) = E[\epsilon_t \epsilon_{t-1}]$. Conditioning on (S_t, S_{t-1}) and using the fact that ϵ_t and ϵ_{t-1} are conditionally independent given their respective scales,

$$E[\epsilon_t \epsilon_{t-1}] = E\big[E[\epsilon_t \epsilon_{t-1} \mid S_t, S_{t-1}]\big] = E\big[E[\epsilon_t \mid S_t]\, E[\epsilon_{t-1} \mid S_{t-1}]\big] = E[0 \cdot 0] = 0.$$

Therefore $\mathrm{cov}(\epsilon_t, \epsilon_{t-1}) = 0$.

Answer - AI

Using the law of iterated expectation:

$$\mathbb{E}[\epsilon_t] = \mathbb{E}\big[\mathbb{E}[\epsilon_t \mid S_t]\big].$$

But

$$\mathbb{E}[\epsilon_t \mid S_t] = 0,$$

so

$$\boxed{\mathbb{E}[\epsilon_t] = 0.}$$

Use the law of total variance:

$$\mathrm{Var}(\epsilon_t) = \mathbb{E}[\mathrm{Var}(\epsilon_t \mid S_t)] + \mathrm{Var}(\mathbb{E}[\epsilon_t \mid S_t]).$$

The second term is zero since the conditional mean is zero.

So:

$$\mathrm{Var}(\epsilon_t) = \mathbb{E}[S_t^2].$$

Now:

$$\mathbb{E}[S_t^2] = \mathrm{Var}(S_t) + (\mathbb{E}[S_t])^2 = \frac{\sigma^2}{1-\alpha^2}.$$

Thus:

$$\boxed{\mathrm{Var}(\epsilon_t) = \frac{\sigma^2}{1-\alpha^2}.}$$

We compute $(\mathrm{Cov}(\epsilon_t, \epsilon_{t-1}))$.

Using iterated expectation:

$$\mathrm{Cov}(\epsilon_t, \epsilon_{t-1}) = \mathbb{E}[\epsilon_t \epsilon_{t-1}]$$

(since both have mean 0).

Now condition on (S_t, S_{t-1}):

$$\mathbb{E}[\epsilon_t \epsilon_{t-1}] = \mathbb{E}\Big[\mathbb{E}[\epsilon_t \epsilon_{t-1} \mid S_t, S_{t-1}]\Big].$$

Given (S_t, S_{t-1}), we have:

- $(\epsilon_t \sim N(0, S_t^2))$
- $(\epsilon_{t-1} \sim N(0, S_{t-1}^2))$
- and they are independent conditional on the volatility process (standard assumption in this type of model).

So:

$$\mathbb{E}[\epsilon_t \epsilon_{t-1} \mid S_t, S_{t-1}] = 0.$$

Therefore:

$$\boxed{\mathrm{Cov}(\epsilon_t, \epsilon_{t-1}) = 0.}$$

By the same argument:

$$\boxed{\mathrm{Cov}(\epsilon_t, \epsilon_{t+1}) = 0.}$$

Evaluation

(c)

Determine the expected values, variances, and one-step covariances of Y_t.

Answer - Me

Since $Y_t = \mu + \epsilon_t$, these follow directly from part (b).

Expected Value is given by

$$E[Y_t] = \mu + E[\epsilon_t] = \mu.$$

Variance is given by

$$\text{var}(Y_t) = \text{var}(\epsilon_t) = \frac{\sigma^2}{1-\alpha^2}.$$

The one-step covariance is given by

$$\text{cov}(Y_t, Y_{t-1}) = \text{cov}(\epsilon_t, \epsilon_{t-1}) = 0.$$

Answer - AI

$$\mathbb{E}[Y_t] = \mathbb{E}[\mu + \epsilon_t] = \mu + \mathbb{E}[\epsilon_t] = \mu.$$

So,

$$\boxed{\mathbb{E}[Y_t] = \mu.}$$

Since (μ) is constant:

$$\text{Var}(Y_t) = \text{Var}(\epsilon_t).$$

Thus,

$$\boxed{\text{Var}(Y_t) = \frac{\sigma^2}{1-\alpha^2}.}$$

We compute:

$$\text{Cov}(Y_t, Y_{t-1}).$$

Substitute:

$$Y_t = \mu + \epsilon_t, \quad Y_{t-1} = \mu + \epsilon_{t-1}.$$

Constants drop out:

$$\text{Cov}(Y_t, Y_{t-1}) = \text{Cov}(\epsilon_t, \epsilon_{t-1}).$$

From the previous part:

$$\text{Cov}(\epsilon_t, \epsilon_{t-1}) = 0.$$

So:

$$\boxed{\mathrm{Cov}(Y_t, Y_{t-1}) = 0.}$$

Similarly:

$$\boxed{\mathrm{Cov}(Y_t, Y_{t+1}) = 0.}$$

Evaluation

(d)

What does this model then imply about a sequence of changes in the index of stock market prices over the given time period?

Note: Just for information (but not really important for this question), an example series of values $Y_1, \ldots, Y_T$ *with* $T = 100$, $\mu = 0$, $\sigma^2 = 1.5$ *and* $\alpha = 0.90$ *is shown in Figure 1.*

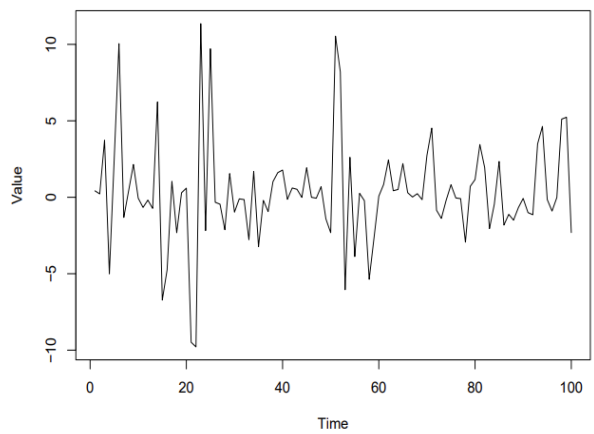

Figure 1: A realization of the model from Additional Question 4.

Answer - Me

The results of parts (b) and (c) together reveal an apparent tension. The Y_t have constant mean μ, constant marginal variance $\sigma^2/(1-\alpha^2)$, and zero one-step covariance — marginally, the sequence of daily changes resembles uncorrelated white noise with no systematic drift.

However, the Y_t are not independent. The variance of Y_t is governed by S_t^2, and S_t follows an AR(1) process with coefficient α. When α is close to 1 (as in the $\alpha = 0.90$ example), the volatility process is highly persistent: a large $|S_t|$ today tends strongly to be followed by a large $|S_{t+1}|$ tomorrow. This induces **volatility clustering** — periods of large swings in the index tend to cluster together in time, as do periods of relative calm, even though successive changes are marginally uncorrelated.

This is consistent with the behavior visible in Figure 1, where episodes of extreme positive and negative changes (around $t = 15$–25 and $t = 50$–60) are concentrated in time rather than distributed uniformly across the series, while other stretches show comparatively little movement. The model therefore captures a key empirical feature of financial markets: the magnitude of daily price changes is predictable in a way that the direction is not.

Answer - AI

This model implies a classic empirical fact of financial data:

Returns are approximately uncorrelated, but volatility is strongly autocorrelated.

So a stock index path under this model:

- looks like random noise in direction
- but with **time-varying intensity ("volatility bursts")**

Evaluation

5.

Consider a formal Goodness of Fit test, such as the Chi-squared test of Karl Pearson. Briefly discuss why Pearson's notion of testing goodness of fit does not fit squarely into either the Fisherian theory of significance tests, or the Neyman-Pearson theory of hypothesis tests.

Answer - Me

Pearson's notion of goodness of fit resists clean placement in either the Fisherian or Neyman-Pearson framework for several related reasons. Most fundamentally, Neyman-Pearson testing requires a null and alternative that together form a logical disjunction, but GOF tests have no such alternative. A theoretical distribution that cannot be rejected is deemed an acceptable graduation of the data, though this is neither acceptance in the Neyman-Pearson behavioral sense nor a claim of uniqueness — multiple theoretical distributions may simultaneously pass.

The direction of evidence is also reversed relative to Fisherian significance testing. Where a small p-value constitutes evidence against the null in Fisher's framework, Pearson's goal is a large p-value; the closer to 1.0, the better the fit. Fisher objected to this explicitly, noting that under a true model, $p = 0.999$ is no more expected than $p = 0.001$. Relatedly, Pearson regarded the posited distribution not as a falsifiable scientific claim but as a mathematical convenience for graduating observed data, which sits uneasily with either school's treatment of the null hypothesis.

Finally, Pearson's interpretation of p-values relied on overlaying a hypothetical random generating mechanism on the data regardless of whether actual random sampling occurred, effectively folding the assumption of random sampling into the hypothesis under test rather than treating it as a precondition. Neither framework accommodates this move naturally, leaving the inferential basis of GOF tests in an uncomfortable middle ground.

Answer - AI

Pearson's chi-squared goodness-of-fit test does not fit squarely into either theory because:

- it lacks a fully specified alternative model (violating Neyman–Pearson structure),
- yet is often used as a decision rule rather than purely an evidential measure (going beyond Fisher's interpretation),
- and fundamentally functions as a **model adequacy diagnostic rather than a strict hypothesis test**.

Evaluation

6.

Consider two continuous, independent random variables X and Y that have distributions with densities $p(x|\theta)$ and $p(y|\lambda)$, respectively, with each having support on the entire real line. Suppose we would like to compute a Monte Carlo approximation to $\Pr(Y > X)$ for given values of θ and λ.

There are two Monte Carlo approximations to $\Pr(Y > X)$ that immediately suggest themselves. Both procedures are based on a sample of values $\{(x_m, y_m) : m = 1, \ldots, M\}$ where the x_m are simulated from $p(x|\theta)$ and the y_m are independently simulated from $p(y|\lambda)$. Approximation A is,

$$P_M^{(A)}(Y > X) = \frac{1}{M} \sum_{m=1}^{M} I(y_m > x_m),$$

where $I(\cdot)$ is the usual indicator function. Approximation B is

$$\overset{(B)}{\underset{M}{\Pr}}(Y > X) = \frac{1}{M} \sum_{m=1}^{M} \frac{1}{M} \sum_{j=1}^{M} I(y_j > x_m).$$

(a)

Identify the explicit integral forms, both equal to $\Pr(Y > X)$, that $P_M^{(A)}$ and $P_M^{(B)}$ are approximating. That is, there are any number of integral expressions that are equal to $\Pr(Y > X)$. You don't get credit for some form just because it is equal to $\Pr(Y > X)$. What we want are those forms that translate directly into the Monte Carlo approximations A and B.

Answer - Me

Both approximations equal $\Pr(Y > X)$ but arise from structurally distinct integral representations.

Approximation A treats (X, Y) as a jointly drawn pair and approximates the single expected value of $I(Y > X)$ with respect to the joint density,

$$\Pr(Y > X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(y > x)\, p(x \mid \theta)\, p(y \mid \lambda)\, dx\, dy,$$

where each simulated pair (x_m, y_m) is drawn together from $p(x|\theta)p(y|\lambda)$ and contributes a single indicator value. This is a direct application of (11.4) with $g(x, y) = I(y > x)$.

Approximation B arises from the iterated expectation $E_X[E_Y[I(Y > X)]]$, in which the inner integral over y is computed first for each fixed x, and the result is then integrated over x,

$$\Pr(Y > X) = \int_{-\infty}^{\infty} \underbrace{\left[\int_{-\infty}^{\infty} I(y > x)\, p(y \mid \lambda)\, dy \right]}_{= \Pr(Y > x)} p(x \mid \theta)\, dx.$$

The inner bracket is $\Pr(Y > x)$ for a fixed value of x, which is itself approximated by a Monte Carlo average over all M simulated y_j values for each x_m. The outer average then integrates over the M simulated x_m values. Approximation B therefore makes use of all M^2 pairs (x_m, y_j) rather than only the M matched pairs used in A.

Answer - AI

Let $(X \sim p(x \mid \theta))$ and $(Y \sim p(y \mid \lambda))$ be independent.

Start from the basic definition:

$$\Pr(Y > X) = \iint I(y > x), p(x \mid \theta), p(y \mid \lambda), dx, dy.$$

Approximation A uses **paired draws** ((x_m, y_m)), so it directly approximates the joint expectation over independent draws of ((X,Y)):

$$\Pr(Y > X) = \mathbb{E}[I(Y > X)] = \int I(y > x), p(x \mid \theta), p(y \mid \lambda), dx, dy.$$

This is the integral form that matches:

$$P_M^{(A)} = \frac{1}{M} \sum_{m=1}^{M} I(y_m > x_m).$$

Approximation B uses **conditioning on (X)** and then averaging over (Y) given (X):

$$\Pr(Y > X) = \int \Pr(Y > x \mid X = x), p(x \mid \theta), dx.$$

Since independence implies ($p(y \mid x) = p(y)$), we get:

$$\Pr(Y > x \mid X = x) = \int I(y > x)\, p(y \mid \lambda)\, dy.$$

So the full expression is:

$$\Pr(Y > X) = \int \left[\int I(y > x)\, p(y \mid \lambda)\, dy \right] p(x \mid \theta) dx.$$

This is exactly what the nested Monte Carlo in B approximates.

Evaluation

(b)

Identify, in terms of expectation operators, again both equal to $\Pr(Y > X)$, the expected values that $P_M^{(A)}$ and $P_M^{(B)}$ are approximating.

Hint: Just because X and Y are independent, that does not preclude us from thinking about the conditional distribution of one given the other. It just means that $p(x|y) = p(x)$, and similarly for Y.

Note About the Question: While these approximations are asymptotically equivalent, there can be differences in Monte Carlo variance for finite Monte Carlo sample sizes, M. In particular, approximation B seems to have a smaller Monte Carlo error than approximation A. Even more interesting, the variances have a nearly constant ratio $V^{(B)}/V^{(A)}$ of about 0.65 to 0.70 as $M \to \infty$ (giving relative efficiency), although both variances are converging to 0, of course.

Answer - Me

Approximation A is directly approximating the single expected value of $I(Y > X)$ with respect to the joint distribution of (X, Y),

$$\Pr(Y > X) = E_{X,Y}[I(Y > X)].$$

Each matched pair (x_m, y_m) is a single draw from the joint distribution, contributing one evaluation of $g(x, y) = I(y > x)$.

Approximation B is approximating the iterated expectation,

$$\Pr(Y > X) = E_X[E_Y[I(Y > X) \mid X]]\,,$$

where the inner expectation $E_Y[I(Y > X) \mid X = x_m] = \Pr(Y > x_m \mid X = x_m)$ is itself approximated by the inner Monte Carlo average over all M draws y_j for each fixed x_m, and the outer expectation over X is then approximated by averaging the M resulting values over all x_m. Although $X \perp Y$ implies $\Pr(Y > x \mid X = x) = \Pr(Y > x)$, the iterated expectation structure is still meaningful and is precisely what gives Approximation B its nested double-sum form. The two forms are equal by the law of total expectation. The iterated form also suggests why B achieves lower Monte Carlo variance than A: by the law of total variance,

$$\mathrm{Var}_{X,Y}(I(Y > X)) = E_X[\mathrm{Var}_Y(I(Y > X) \mid X)] + \mathrm{Var}_X(E_Y[I(Y > X) \mid X]),$$

so the quantity Approximation B averages over x_m — namely $E_Y[I(Y > X) \mid X]$ — has variance equal only to the second term, strictly less than the full $\mathrm{Var}_{X,Y}(I(Y > X))$ driving the variance of A. This is a Rao-Blackwell type argument: conditioning on X yields a smoother, lower-variance integrand.

Answer - AI

For Approximation A, This is a **single expectation over the joint distribution**:

$$\Pr(Y > X) = \mathbb{E}\big[I(Y > X)\big],$$

where $((X, Y) \sim p(x \mid \theta)p(y \mid \lambda))$.

So $(P_M^{(A)})$ is estimating:

$$\mathbb{E}[I(Y > X)].$$

For Approximation B, rewrite using iterated expectation:

$$\Pr(Y > X) = \mathbb{E}_X\left[\mathbb{E}_Y[I(Y > X) \mid X]\right].$$

Since independence implies the inner expectation does not depend on conditioning beyond (X):

$$\mathbb{E}_Y[I(Y > X) \mid X] = \Pr(Y > X \mid X).$$

So the target becomes:

$$\Pr(Y > X) = \mathbb{E}\left[\Pr(Y > X \mid X)\right].$$

Thus $(P_M^{(B)})$ approximates:

$$\mathbb{E}_X\left[\mathbb{E}_Y[I(Y > X) \mid X]\right].$$

Key conceptual difference (why B can have lower variance)

- **A:** estimates a *single noisy indicator per pair* $(I(Y > X))$.
- **B:** replaces the inner randomness in (Y) with its *conditional average given (X)*, reducing Monte Carlo noise via Rao–Blackwellization.

So B is effectively a **variance-reduced (Rao–Blackwellized) version** of A, which explains the consistently lower asymptotic variance.

Evaluation